

MATH 2400 - Midterm 1

In everything that follows, $\vec{r} = \vec{r}(t) = \langle x, y, z \rangle$ is the variable position vector and $\vec{r}_0 = \vec{r}(t_0) = \langle x_0, y_0, z_0 \rangle$ is a constant vector. The expressions below are written for vectors with three components. There are corresponding formulas for vectors with two components (except for the cross product) with the third component deleted.

1 12.1 - 3D Coordinate Systems

- Rectangular coordinate system (x, y, z) . Coordinate axes (x -axis, y -axis, z -axis) and coordinate planes (xy -plane, xz -plane, yz -plane).

2 12.2 - Vectors

- Vector arithmetic: $\vec{v} + \vec{w}$, $\vec{v} - \vec{w}$, $c\vec{v}$, both geometrically (arrows) and algebraically (component-wise).
- The vector between points $P = (x_1, y_1, z_1)$ and $Q = (x_2, y_2, z_2)$ is given by the difference of the coordinates: $\overrightarrow{PQ} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$.
- Norm (magnitude) of a vector: $|\vec{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2}$.
- If $|\vec{v}| \neq 0$, then $\vec{u} = \frac{1}{|\vec{v}|}\vec{v}$ is **normalization** of $\vec{v}$. This is a unit vector ($|\vec{u}| = 1$) in the direction of $\vec{v}$.
- The vectors $\vec{v}$ and $\vec{w}$ are parallel if there is a non-zero k with $\vec{w} = k\vec{v}$.

3 12.3 - Dot Product

- $\vec{v} \cdot \vec{w} = v_1w_1 + v_2w_2 + v_3w_3$
- $|\vec{v}|^2 = \vec{v} \cdot \vec{v}$
- If $\vec{v}$ and $\vec{w}$ are not zero vectors, then the angle between them, θ , is given by $\cos(\theta) = \frac{\vec{v} \cdot \vec{w}}{|\vec{v}||\vec{w}|}$.
- If $\vec{v} \cdot \vec{w} = 0$ then the vectors are orthogonal.

4 12.4 - Cross Product

- $\vec{v} \times \vec{w} = \det \begin{pmatrix} \vec{i} & \vec{j} & \vec{k} \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{pmatrix}$
- $\vec{v} \times \vec{w} = -\vec{w} \times \vec{v}$
- $\vec{v} \times \vec{w}$ is orthogonal to both $\vec{v}$ and $\vec{w}$ and its direction is determined by the **right-hand rule**.
- $|\vec{v} \times \vec{w}|$ is the area of the parallelogram spanned by vectors $\vec{v}$ and $\vec{w}$.

5 12.5 - Lines and Planes

5.1 Lines

- The line through $\vec{r}_0$ in the direction of $\vec{v}$ is given by $\vec{r}(t) = \vec{r}_0 + t\vec{v}$.

5.2 Planes

- There is a unique plane through any three non-collinear points, P, Q, R . Find the vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$, then find $\vec{n} = \overrightarrow{PQ} \times \overrightarrow{PR}$. This is the normal to the plane. Take P as the point that the plane passes through.
- There is a unique plane containing any line, L , and point, P , not on the line. Find a vector from any point on the line and the point P , take the direction vector of the line, and use those two vectors in the same way as in the previous case.
- There is a unique plane containing any two distinct parallel lines. Find any vector from a point on one line to any point on the other, and proceed as in the previous case.
- Vector equation of plane: $\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0$
- Linear equation of plane: $ax + by + cz = d$
- The normal to the plane $ax + by + cz = d$ is $\vec{n} = \langle a, b, c \rangle$.

6 12.6 - Functions and Surfaces

- We work primarily with functions of the form $F(x, y, z) = k$. Setting any of the variables equal to a constant, for example $z = z_0$, give **traces**. These are curves in the plane determined by the equation of the constant variable, which in the example would be the plane $z = z_0$.
- Graphing multiple traces (with the same variable being set equal to different constants) on the same set of axes (these will involve the two variables that aren't set to a constant) gives a **contour plot**.
- If one (or more) of the variables is missing from your function, then you have a cylindrical surface (not to be confused with a cylinder, even though cylindrical surfaces are also sometimes just called cylinders (it's confusing, I know)). For example, $y = x^2$ considered as a function of two variables (x and y) is a parabola, but considered as a function of three variables (x, y , and z) is a parabolic cylinder (imagine a parabola in the xy -plane slid along the z -axis).
- There are six **Quadric Surfaces**:
 1. Ellipsoid: $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$
 2. Elliptical Paraboloid: $\frac{z}{c} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$
 3. Hyperbolic Paraboloid: $\frac{z}{c} = \frac{x^2}{a^2} - \frac{y^2}{b^2}$
 4. Cone: $\frac{z^2}{c^2} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$
 5. Hyperboloid of One Sheet: $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$
 6. Hyperboloid of Two Sheets: $-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$
- The traces of all of these surfaces are ellipses, parabolas, or hyperbolas (with the degenerate case of lines with the cone).
- Changing the ordering of the variables with respect to the first-power term (with the paraboloids) or the minus signs (with the cone and hyperboloids) will change the axis that the surfaces is oriented to.
- Changing the values of a, b , and c will change the "size" of the shapes of the surfaces.

7 15.7 and 15.8 - Cylindrical and Spherical Coordinates

7.1 Cylindrical Coordinates

Given a point (x, y, z) in rectangular coordinates, project this down to the point $(x, y, 0)$ in the xy -plane. Let r be the distance from the point $(x, y, 0)$ from the origin and let θ be the angle that the vector $\langle x, y, 0 \rangle$ makes with the positive x -axis.

- $x = r \cos(\theta), y = r \sin(\theta), z = z$.

7.2 Spherical Coordinates

Given (x, y, z) in rectangular coordinates, let ρ be the distance from this point to the origin, θ the angle that the vector $\langle x, y, 0 \rangle$ makes with the positive x -axis, and ϕ the angle that $\langle x, y, z \rangle$ makes with the positive z -axis.

- $x = \rho \cos(\theta) \sin(\phi), y = \rho \sin(\theta) \sin(\phi), z = \rho \cos(\phi)$.

8 13.1 - Vector Functions and Space Curves

- A **vector-valued function** (or just **vector function**) has the form $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$.
- The graph of a vector function is a curve in 3D space, called a **space curve**.
- The **parametric equations** of a space curve are just the component functions of the curve:
 $x = f(t), y = g(t), z = h(t)$.

9 13.2 - Derivatives and Integrals of Vector Functions

- The derivative of a vector function is just the vector function of the derivatives of the component functions:
 $\vec{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle$.
- The derivative of a vector function at a point on a space curve gives the tangent vector to the curve at the point: $\vec{r}'(t_0) = \langle f'(t_0), g'(t_0), h'(t_0) \rangle$.
- The **unit tangent vector** is the normalized tangent vector: $T(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}$.
- The tangent line to a space curve at a point $\vec{r}(t_0)$ is given by $\vec{r}(t) = \vec{r}(t_0) + t\vec{r}'(t_0)$.
- The definite integral of a vector function is the vector of definite integrals of the component functions:
 $\int_a^b \vec{r}(t) dt = \langle \int_a^b f(t) dt, \int_a^b g(t) dt, \int_a^b h(t) dt \rangle$.

10 13.3 - Arc Length

Let C be a curve parameterized by $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$.

- The **arc length** of the portion of C between $a \leq t \leq b$ is

$$L = \int_a^b |\vec{r}'(t)| dt$$

- This can be interpreted as “summing” the speed that the particle is moving at a given moment in time, t , from time $t = a$ to $t = b$.
- The arc length parameterization is given by

$$s = \int_a^t |\vec{r}'(u)| du = g(t) \Rightarrow t = g^{-1}(s)$$

Then change parameters from t to s : $\vec{r}(t) = \vec{r}(g^{-1}(s))$.

- Using the arc length parameter, each increase of s by one unit moves you exactly one unit along the curve C .

11 16.6 - Parametric Surfaces

- Parametric surfaces are given by $\vec{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle$.
- There are four main types of parametric surfaces that we will work with:

1. Surfaces of the form $z = f(x, y)$ (the roles of the independent and dependent variables can be changed). These can be parameterized by $\vec{r}(x, y) = \langle x, y, f(x, y) \rangle$.
 2. Surfaces in Cylindrical Coordinates. These are cylinders or portions of cylinders. Example: $\vec{r}(\theta, z) = \langle 2 \cos(\theta), 2 \sin(\theta), z \rangle$ is the cylinder of radius 2 opening along the z -axis. Changing the “free” variable (z , in the example) will change the axis that the cylinder is parallel to.
 3. Surfaces in Spherical Coordinates. There are spheres or portions of spheres. Example: $\vec{r}(\theta, \phi) = \langle 2 \cos(\theta) \sin(\phi), 2 \sin(\theta) \sin(\phi), 2 \cos(\phi) \rangle$ is the sphere of radius 2 centered at the origin.
 4. Surfaces of Revolution. If $y = f(x)$ is rotated about the x -axis, the surface of revolution has parameterization $\vec{r}(x, \theta) = \langle x, f(x) \cos(\theta), f(x) \sin(\theta) \rangle$. If $x = g(y)$ is rotated about the y -axis, the surface of revolution has parameterization $\vec{r}(y, \theta) = \langle g(y) \cos(\theta), y, g(y) \sin(\theta) \rangle$.
- If you restrict the values of the parameter, $a \leq u \leq b, c \leq v \leq d$, you get a portion of the surface. You are basically mapping the rectangle in the uv -plane defined by the inequalities to a portion of the surface in xyz -space.

12 General Comments

- Try to relate the more complicated things that you are asked to work with back to simpler things that you are already familiar with. This is one of the main themes of Calc 3. Instead of trying to visualize a surface in its entirety, try to visualize the traces. Things like that.
- Most of the geometric intuition that you built up in Calc 1 and 2 goes through to the concepts you are working with in Calc 3. The derivative of a vector-valued function is telling you about rate of change, just like derivatives did in Calc 1. Definite integrals “measure” things (like arc length), just like in Calc 1 and 2.
- This exam will be mostly about geometric concepts. Use your geometric intuition wherever possible, keeping in mind that things are in general “bigger” in the multi-variable setting. What’s a plane? A higher dimensional line. What’s a sphere? A higher dimensional circle. And so on.
- I would encourage you to work on the free response questions first. They are generally harder and take more time than the multiple choice questions. It’s better to feel rushed when you’re working with something easier than when you’re trying to do something more complicated.
- I would also encourage you to start with a problem that you feel very confident in your ability to do. There is nothing more demoralizing than spending twenty minutes on something and getting nowhere.
- Please keep in mind that we are only going to ask you questions that we feel you should be able to do. This means that if something seems impossibly hard, you’re probably thinking about it the wrong way or trying to do something that you shouldn’t be doing. If you find yourself in this situation, take a deep breath and start over and try something different.
- Take some time to look back over your answers. Ask if they make sense (for the multiple choice) and if you can follow your own work (for the free response). If things can be made better, and you have the time, make them better. Find the balance between being critical of your own work and doubting yourself.
- Remember, presentation and communication matter. Write things up clearly and coherently. Use full English sentences. Don’t have a bunch of arrows and things crossed-out and this thing boxed and then that thing boxed, etc.
- Watch the little things, like minus signs. Be careful with distributing and fractions. Use parentheses.